

Elliot Uvero

Student #: 2611467

EE 446 Assignment 1

Problem 1 - Part(e):

In my notebook, all the models were trained using the default ADAM optimizer without any early stopping callbacks. The base model was trained using 20 epochs and the pruned and student models were trained using 10 epochs. The wine dataset consisted of only 178 data points and these were separated using a train-test split (70% train, 30% test) with `random_state=42` and `stratify=y` to ensure all three classes were equally distributed in both the training and test datasets. Below are the overall accuracies and model sizes for the base model and all the other models generated and tested in parts (b), (c) and (d).

Note: I sometimes got different results when re-running my notebook. This is perhaps due to the very small dataset size and the effects of random sampling in the test-train split.

	Model	Test Accuracy	Model Size (KB)
0	BASE	0.981481	14.062500
1	FLOAT 16	0.981481	8.937500
2	DYNAMIC	0.981481	8.164062
3	INT8	0.981481	5.734375
4	PRUNED	0.907407	8.187500
5	KD	0.925926	6.089844

Figure 1: Model Test Accuracies and Sizes

The best performing model was the INT8 model which had a 0.9815 accuracy and a 5.73 KB size.

I explored three different approaches.

- *INT8 Quantization of Pruned Model*
- *INT8 Quantization of Student Model*
- *Pruning and INT8 of Student Model*

None of the models performed as well as the INT8 quantized base model. The INT8 quantized pruned model was 5.75 KB and both INT8 quantized student pruned and unpruned models were 3.61 KB. Results are summarized in the following figures.

```

INT8 TFLite Model Test Accuracy: 0.9074
INT8 TFLite model size: 5.75 KB

```

	precision	recall	f1-score	support
0	1.0000	1.0000	1.0000	18
1	0.8077	1.0000	0.8936	21
2	1.0000	0.6667	0.8000	15
accuracy			0.9074	54
macro avg	0.9359	0.8889	0.8979	54
weighted avg	0.9252	0.9074	0.9031	54

```

2026-07-24 16:16:35.900499: W tensorflow/compiler/mlir/lite/python/tf_
2026-07-24 16:16:35.900508: W tensorflow/compiler/mlir/lite/python/tf_
2026-07-24 16:16:35.900577: I tensorflow/cc/saved_model/reader.cc:83]
2026-07-24 16:16:35.900826: I tensorflow/cc/saved_model/reader.cc:51]
2026-07-24 16:16:35.900830: I tensorflow/cc/saved_model/reader.cc:146]
2026-07-24 16:16:35.901421: I tensorflow/cc/saved_model/loader.cc:233]
2026-07-24 16:16:35.906910: I tensorflow/cc/saved_model/loader.cc:217]
2026-07-24 16:16:35.909168: I tensorflow/cc/saved_model/loader.cc:316]
fully_quantize: 0, inference_type: 6, input_inference_type: INT8, outp

```

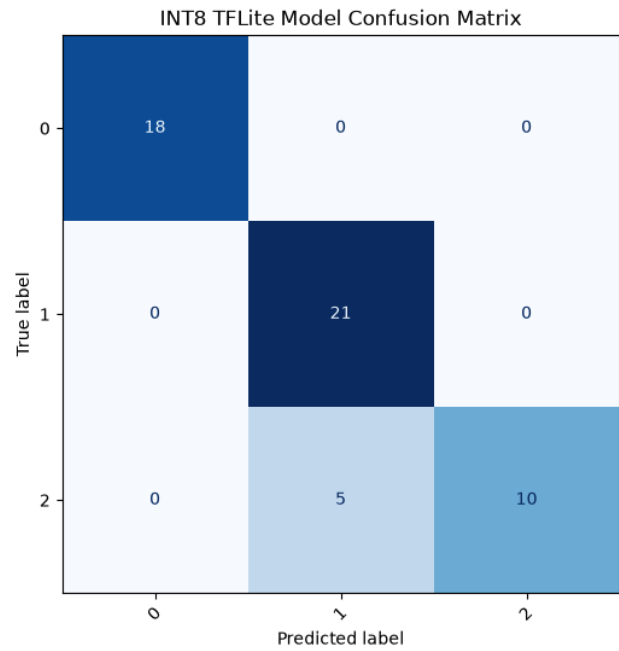


Figure 2: INT8 Quantized Pruned Model Performance and Size

INT8 TFLite Model Test Accuracy: 0.9259
 INT8 TFLite model size: 3.61 KB

	precision	recall	f1-score	support
0	1.0000	1.0000	1.0000	18
1	0.9048	0.9048	0.9048	21
2	0.8667	0.8667	0.8667	15
accuracy			0.9259	54
macro avg	0.9238	0.9238	0.9238	54
weighted avg	0.9259	0.9259	0.9259	54

```

2026-07-24 16:16:36.105140: I tensorflow/cc/saved_model/reader.cc:83] Rea
2026-07-24 16:16:36.105496: I tensorflow/cc/saved_model/reader.cc:51] Rea
2026-07-24 16:16:36.105500: I tensorflow/cc/saved_model/reader.cc:146] Re
2026-07-24 16:16:36.106414: I tensorflow/cc/saved_model/loader.cc:233] Re
2026-07-24 16:16:36.118889: I tensorflow/cc/saved_model/loader.cc:217] Ru
2026-07-24 16:16:36.122292: I tensorflow/cc/saved_model/loader.cc:316] Sa
fully_quantize: 0, inference_type: 6, input_inference_type: INT8, output_
  
```

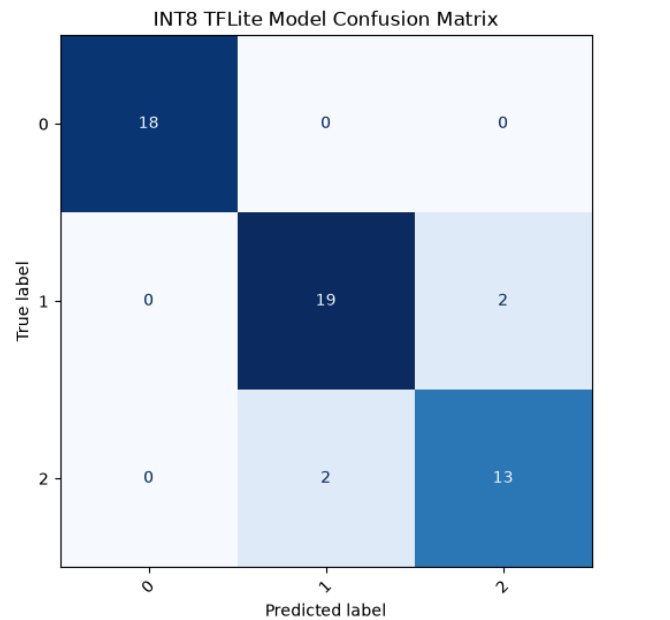


Figure 3: INT8 Quantized Student Model Performance and Size

INT8 TFLite Model Test Accuracy: 0.9074
 INT8 TFLite model size: 3.61 KB

	precision	recall	f1-score	support
0	0.9474	1.0000	0.9730	18
1	0.9000	0.8571	0.8780	21
2	0.8667	0.8667	0.8667	15
accuracy			0.9074	54
macro avg	0.9047	0.9079	0.9059	54
weighted avg	0.9065	0.9074	0.9065	54

```

/Users/eauvero/ai/projects/tinym1-arduino/lib/python3.11/site-packages/ter
warnings.warn(
2026-07-24 16:16:36.813553: W tensorflow/compiler/mlir/lite/python/tf_tfl_
2026-07-24 16:16:36.813562: W tensorflow/compiler/mlir/lite/python/tf_tfl_
2026-07-24 16:16:36.813642: I tensorflow/cc/saved_model/reader.cc:83] Read
2026-07-24 16:16:36.813838: I tensorflow/cc/saved_model/reader.cc:51] Read
2026-07-24 16:16:36.813844: I tensorflow/cc/saved_model/reader.cc:146] Rea
2026-07-24 16:16:36.814356: I tensorflow/cc/saved_model/loader.cc:233] Res
2026-07-24 16:16:36.819448: I tensorflow/cc/saved_model/loader.cc:217] Run
2026-07-24 16:16:36.821495: I tensorflow/cc/saved_model/loader.cc:316] Sav
fully_quantize: 0, inference_type: 6, input_inference_type: INT8, output_i

```

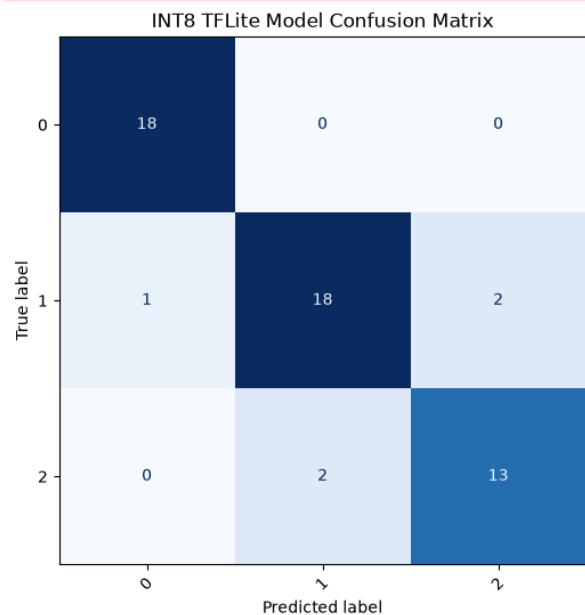


Figure 4: INT8 Quantized Pruned Student Model Performance and Size

Problem 2.1:

The following learning blocks are available

- *Classification*: This block supports supervised learning applications that learn patterns from data and predict categories. One example would be an application to detect if a driver is distracted.
- *Regression*: This block supports supervised learning applications that learn patterns from data and predict numeric values from patterns such as future power usage given the temperature and length of day.
- *Anomaly Detection*: There are GMM and K-Means versions available. This block supports unsupervised learning applications. One example could be fraud detection or security alert systems.

Problem 2.2:

The following layers are available

- *Dense*: Fully connected layer used for processed data. Used in feed-forward neural networks.
- *1D Convolution / pooling*: Layer used in Convolutional Neural Networks that learns features while taking spatial information into account along a single dimension. This can be used on raw data such as images or spectrograms.
- *2D Convolution / pooling*: Layer used in Convolutional Neural Networks that learns features while taking spatial information into account along two dimensions.
- *Reshape*: This can transform one-dimensional data into multi-dimensional data that can be used as an input to a convolutional layer. This is useful for deep learning models on raw data.
- *Flatten*: This layer squashes multi-dimensional data into a single dimension. This is needed in transforming multi-dimensional data from convolutional layers into something that can be fed into a standard Dense layer.
- *Dropout*: This layer reduces the risk of overfitting a model by randomly cutting a fraction of network connections between dense layers during training.

Problem 2.3:

There is only one Quantized (int8) option under the Model Optimizations and Performance panel in the Deployment section.

Problem 2.4:

The free-tier version of Edge Impulse allows users to generate synthetic image and audio data. For images, a user can input a prompt to either the Black Forest Labs Flux-Pro or OpenAI's GPT-Image-2 Synthetic Image Generator models to generate synthetic images based on the prompt. For audio, a user can use OpenAI's text-to-speech model to generate synthetic voice samples. Machine learning models are heavily dependent on the volume and quality of data used in training them. Data acquisition, maintenance and sanitation can be difficult and expensive in practice. There can be issues with privacy protection and compliance. Other times, data for edge cases can be very scarce. Sometimes datasets for classification applications may overrepresent a certain category which could bias the model predictions. Generating and using synthetic data for machine learning applications helps overcome some of these challenges.